Module IV: Optimization of Inequality Constrained Continuous Functions

Optimization Techniques (24EST372)

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Outline

Objectives

- Understand constrained optimization problems with inequality constraints.
- Learn the Lagrangean method for inequality-constrained problems.
- Master the Karush-Kuhn-Tucker (KKT) conditions for finding stationary points.
- Apply these methods to theoretical and AI-related optimization problems.

Constrained Problems with Inequality Constraints

Optimize a continuous function subject to inequality constraints:

$$\text{Maximize/Minimize } f(\mathbf{x}), \quad \mathbf{x} = (x_1, x_2, \ldots, x_n)^T$$
$$\text{Subject to } g_i(\mathbf{x}) \leq 0, \quad i = 1, \ldots, m$$

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$$\text{Subject to } g_i(\mathbf{x}) \leq 0, \quad i = 1, \ldots, m$$

Goal: Find $\mathbf{x}^*$ that optimizes $f(\mathbf{x})$ while satisfying $g_i(\mathbf{x}) \leq 0$.

Lagrangean Method for Inequality Constraints

Extends Lagrange multipliers to handle inequality constraints $g_i(\mathbf{x}) \leq 0$.

Form the Lagrangian: $\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{i=1}^{m} \mu_i g_i(\mathbf{x})$, where $\mu_i \geq 0$.

Solutions must satisfy KKT conditions, accounting for active ($g_i(\mathbf{x}) = 0$) and inactive ($g_i(\mathbf{x}) < 0$) constraints.

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Solutions must satisfy KKT conditions, accounting for active ($g_i(\mathbf{x}) = 0$) and inactive ($g_i(\mathbf{x}) < 0$) constraints.

Key: $\mu_i = 0$ for inactive constraints due to complementary slackness.

KKT Necessary Conditions

For a minimization problem: Minimize $f(\mathbf{x})$ subject to $g_i(\mathbf{x}) \leq 0$, the KKT conditions are:

1. Stationarity: $\nabla f(\mathbf{x}) + \sum_{i=1}^{m} \mu_i \nabla g_i(\mathbf{x}) = \mathbf{0}$
2. Primal Feasibility: $g_i(\mathbf{x}) \leq 0, \quad i = 1, \ldots, m$
3. Dual Feasibility: $\mu_i \geq 0, \quad i = 1, \ldots, m$
4. Complementary Slackness: $\mu_i g_i(\mathbf{x}) = 0, \quad i = 1, \ldots, m$

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3. Dual Feasibility: $\mu_i \geq 0, \quad i = 1, \ldots, m$
4. Complementary Slackness: $\mu_i g_i(\mathbf{x}) = 0, \quad i = 1, \ldots, m$

Solutions satisfying these conditions are stationary points (candidates for optima).

Step-by-Step Procedure for KKT Conditions

To solve: Minimize $f(\mathbf{x})$ subject to $g_i(\mathbf{x}) \leq 0$:

Formulate Lagrangian: $\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{i=1}^{m} \mu_i g_i(\mathbf{x})$.

Write KKT Conditions:

$$\begin{cases} \nabla_{\mathbf{x}} \mathcal{L} = \nabla f(\mathbf{x}) + \sum_{i=1}^{m} \mu_i \nabla g_i(\mathbf{x}) = \mathbf{0} \\ g_i(\mathbf{x}) \leq 0 \\ \mu_i \geq 0 \\ \mu_i g_i(\mathbf{x}) = 0 \end{cases}$$

Identify Active Constraints: Assume $g_i(\mathbf{x}) = 0$ (active) or $\mu_i = 0$ (inactive).

Solve System: Solve for $\mathbf{x}$ and $\boldsymbol{\mu}$.

Classify Solutions: Use second-order conditions (e.g., Hessian on active constraints) or evaluate $f(\mathbf{x})$.

Verify Feasibility: Ensure all constraints are satisfied.

Solved Problem 1: Standard Optimization

Minimize $f(x, y) = x^2 + y^2$ subject to:

$$g(x, y) = x + y - 4 \leq 0$$

Step 1: Form Lagrangian

$$\mathcal{L}(x, y, \mu) = x^2 + y^2 + \mu(x + y - 4)$$

Step 2: Write KKT Conditions

Stationarity: $\frac{\partial \mathcal{L}}{\partial x} = 2x + \mu = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = 2y + \mu = 0$

Primal Feasibility: $x + y - 4 \leq 0$

Dual Feasibility: $\mu \geq 0$

Complementary Slackness: $\mu(x + y - 4) = 0$

Solved Problem 1: Solution

Step 3: Solve KKT Conditions From $\frac{\partial \mathcal{L}}{\partial x} = 2x + \mu = 0$: $\mu = -2x$ From $\frac{\partial \mathcal{L}}{\partial y} = 2y + \mu = 0$: $\mu = -2y$ Equate: $-2x = -2y \implies x = y$

Solved Problem 1: Solution

Step 3: Solve KKT Conditions From $\frac{\partial \mathcal{L}}{\partial x} = 2x + \mu = 0$: $\mu = -2x$ From $\frac{\partial \mathcal{L}}{\partial y} = 2y + \mu = 0$: $\mu = -2y$ Equate: $-2x = -2y \implies x = y$
Case 1: $\mu = 0$ **(Inactive Constraint)**

$$\mu = 0 \implies x = 0, y = 0$$

Check: $x + y = 0 + 0 = 0 \leq 4$ (feasible) Objective: $f(0, 0) = 0$

Solved Problem 1: Solution

Step 3: Solve KKT Conditions From $\frac{\partial \mathcal{L}}{\partial x} = 2x + \mu = 0$: $\mu = -2x$ From $\frac{\partial \mathcal{L}}{\partial y} = 2y + \mu = 0$: $\mu = -2y$ Equate: $-2x = -2y \implies x = y$
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$$\mu = 0 \implies x = 0, y = 0$$

Check: $x + y = 0 + 0 = 0 \leq 4$ (feasible) Objective: $f(0,0) = 0$
Case 2: $\mu \neq 0$ **(Active Constraint,** $x + y = 4$**)**

$$x = y, \quad x + x = 4 \implies 2x = 4 \implies x = 2, y = 2$$

$\mu = -2x = -2 \cdot 2 = -4$, but $\mu \geq 0$, so infeasible.

Solved Problem 1: Solution

Step 3: Solve KKT Conditions From $\frac{\partial \mathcal{L}}{\partial x} = 2x + \mu = 0$: $\mu = -2x$ From $\frac{\partial \mathcal{L}}{\partial y} = 2y + \mu = 0$: $\mu = -2y$ Equate: $-2x = -2y \implies x = y$
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$$x = y, \quad x + x = 4 \implies 2x = 4 \implies x = 2, y = 2$$

$\mu = -2x = -2 \cdot 2 = -4$, but $\mu \geq 0$, so infeasible.
Step 4: Classify Hessian of f: $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, positive definite. Only feasible solution: $(0,0)$, $f = 0$. **Solution**: Minimum at $(x,y) = (0,0)$, $f = 0$.

Solved Problem 2: Neural Network Optimization

Minimize $f(w_1, w_2) = w_1^2 + w_2^2 + w_1 w_2$ (regularized loss) subject to:

$$g(w_1, w_2) = w_1 + w_2 - 3 \leq 0$$

Step 1: Form Lagrangian

$$\mathcal{L}(w_1, w_2, \mu) = w_1^2 + w_2^2 + w_1 w_2 + \mu(w_1 + w_2 - 3)$$

Step 2: Write KKT Conditions

Stationarity: $\dfrac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + w_2 + \mu = 0, \quad \dfrac{\partial \mathcal{L}}{\partial w_2} = w_1 + 2w_2 + \mu = 0$

Primal Feasibility: $w_1 + w_2 - 3 \leq 0$

Dual Feasibility: $\mu \geq 0$

Complementary Slackness: $\mu(w_1 + w_2 - 3) = 0$

Solved Problem 2: Solution

Step 3: Solve KKT Conditions From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + w_2 + \mu = 0$: $w_2 = -2w_1 - \mu$
From $\frac{\partial \mathcal{L}}{\partial w_2} = w_1 + 2w_2 + \mu = 0$: $w_1 + 2w_2 + \mu = 0$

Solved Problem 2: Solution

Step 3: Solve KKT Conditions From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + w_2 + \mu = 0$: $w_2 = -2w_1 - \mu$
From $\frac{\partial \mathcal{L}}{\partial w_2} = w_1 + 2w_2 + \mu = 0$: $w_1 + 2w_2 + \mu = 0$
Case 1: $\mu = 0$ **(Inactive Constraint)**

$$w_2 = -2w_1, \quad w_1 + 2(-2w_1) = w_1 - 4w_1 = -3w_1 = 0 \implies w_1 = 0, w_2 = 0$$

Check: $w_1 + w_2 = 0 + 0 = 0 \leq 3$ (feasible) Objective: $f(0, 0) = 0$

Solved Problem 2: Solution

Step 3: Solve KKT Conditions From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + w_2 + \mu = 0$: $w_2 = -2w_1 - \mu$
From $\frac{\partial \mathcal{L}}{\partial w_2} = w_1 + 2w_2 + \mu = 0$: $w_1 + 2w_2 + \mu = 0$
Case 1: $\mu = 0$ **(Inactive Constraint)**

$$w_2 = -2w_1, \quad w_1 + 2(-2w_1) = w_1 - 4w_1 = -3w_1 = 0 \implies w_1 = 0, w_2 = 0$$

Check: $w_1 + w_2 = 0 + 0 = 0 \leq 3$ (feasible) Objective: $f(0,0) = 0$
Case 2: $\mu \neq 0$ **(Active Constraint,** $w_1 + w_2 = 3$**)** Substitute $w_2 = 3 - w_1$ into $w_2 = -2w_1 - \mu$:

$$3 - w_1 = -2w_1 - \mu \implies \mu = w_1 + 3$$

Into second equation: $w_1 + 2(3 - w_1) + (w_1 + 3) = w_1 + 6 - 2w_1 + w_1 + 3 = 9 = 0$ (infeasible).

Solved Problem 2: Solution

Step 3: Solve KKT Conditions From $\frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + w_2 + \mu = 0$: $w_2 = -2w_1 - \mu$
From $\frac{\partial \mathcal{L}}{\partial w_2} = w_1 + 2w_2 + \mu = 0$: $w_1 + 2w_2 + \mu = 0$
Case 1: $\mu = 0$ **(Inactive Constraint)**

$$w_2 = -2w_1, \quad w_1 + 2(-2w_1) = w_1 - 4w_1 = -3w_1 = 0 \implies w_1 = 0, w_2 = 0$$

Check: $w_1 + w_2 = 0 + 0 = 0 \leq 3$ (feasible) Objective: $f(0,0) = 0$
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Into second equation: $w_1 + 2(3 - w_1) + (w_1 + 3) = w_1 + 6 - 2w_1 + w_1 + 3 = 9 = 0$ (infeasible).

Step 4: Classify Hessian of f: $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, det $= 2 \cdot 2 - 1 \cdot 1 = 3 > 0$, $\partial^2 f / \partial w_1^2 = 2 > 0$, positive definite. Only feasible solution: $(0,0)$, $f = 0$. **Solution**: Minimum at $(w_1, w_2) = (0,0)$, $f = 0$. **Interpretation**: Optimal weights minimize loss within the constraint.

Solved Problem 3: Quadratic Objective with Two Constraints

Minimize $f(x, y) = x^2 + y^2$ subject to:

$$g_1(x, y) = x - 2 \leq 0, \quad g_2(x, y) = y - 3 \leq 0$$

Solved Problem 3: Quadratic Objective with Two Constraints

Minimize $f(x, y) = x^2 + y^2$ subject to:

$$g_1(x, y) = x - 2 \leq 0, \quad g_2(x, y) = y - 3 \leq 0$$

Step 1: Form Lagrangian

$$\mathcal{L}(x, y, \mu_1, \mu_2) = x^2 + y^2 + \mu_1(x - 2) + \mu_2(y - 3)$$

Step 2: Write KKT Conditions

Stationarity: $\dfrac{\partial \mathcal{L}}{\partial x} = 2x + \mu_1 = 0, \quad \dfrac{\partial \mathcal{L}}{\partial y} = 2y + \mu_2 = 0$

Primal Feasibility: $x - 2 \leq 0, \quad y - 3 \leq 0$

Dual Feasibility: $\mu_1 \geq 0, \quad \mu_2 \geq 0$

Complementary Slackness: $\mu_1(x - 2) = 0, \quad \mu_2(y - 3) = 0$

Solved Problem 3: Solution

Step 3: Solve KKT Conditions From stationarity: $\mu_1 = -2x$, $\mu_2 = -2y$.

Solved Problem 3: Solution

Step 3: Solve KKT Conditions From stationarity: $\mu_1 = -2x$, $\mu_2 = -2y$.
Case 1: $\mu_1 = 0, \mu_2 = 0$

$$x = 0, \quad y = 0$$

Check: $x - 2 = 0 - 2 = -2 \leq 0$, $y - 3 = 0 - 3 = -3 \leq 0$ (feasible). Objective: $f(0,0) = 0$.

Solved Problem 3: Solution

Step 3: Solve KKT Conditions From stationarity: $\mu_1 = -2x$, $\mu_2 = -2y$.
Case 1: $\mu_1 = 0, \mu_2 = 0$

$$x = 0, \quad y = 0$$

Check: $x - 2 = 0 - 2 = -2 \leq 0$, $y - 3 = 0 - 3 = -3 \leq 0$ (feasible). Objective: $f(0,0) = 0$.
Case 2: $\mu_1 = 0, \mu_2 \neq 0 \implies y = 3$

$$x = 0, \quad y = 3, \quad \mu_2 = -2 \cdot 3 = -6 \text{ (infeasible, since } \mu_2 \geq 0).$$

Solved Problem 3: Solution

Step 3: Solve KKT Conditions From stationarity: $\mu_1 = -2x$, $\mu_2 = -2y$.
Case 1: $\mu_1 = 0, \mu_2 = 0$

$$x = 0, \quad y = 0$$

Check: $x - 2 = 0 - 2 = -2 \leq 0$, $y - 3 = 0 - 3 = -3 \leq 0$ (feasible). Objective: $f(0, 0) = 0$.
Case 2: $\mu_1 = 0, \mu_2 \neq 0 \implies y = 3$

$$x = 0, \quad y = 3, \quad \mu_2 = -2 \cdot 3 = -6 \text{ (infeasible, since } \mu_2 \geq 0\text{).}$$

Case 3: $\mu_1 \neq 0, \mu_2 = 0 \implies x = 2$

$$x = 2, \quad y = 0, \quad \mu_1 = -2 \cdot 2 = -4 \text{ (infeasible).}$$

Solved Problem 3: Solution

Step 3: Solve KKT Conditions From stationarity: $\mu_1 = -2x$, $\mu_2 = -2y$.
Case 1: $\mu_1 = 0, \mu_2 = 0$

$$x = 0, \quad y = 0$$

Check: $x - 2 = 0 - 2 = -2 \leq 0$, $y - 3 = 0 - 3 = -3 \leq 0$ (feasible). Objective: $f(0, 0) = 0$.
Case 2: $\mu_1 = 0, \mu_2 \neq 0 \implies y = 3$

$$x = 0, \quad y = 3, \quad \mu_2 = -2 \cdot 3 = -6 \text{ (infeasible, since } \mu_2 \geq 0\text{).}$$

Case 3: $\mu_1 \neq 0, \mu_2 = 0 \implies x = 2$

$$x = 2, \quad y = 0, \quad \mu_1 = -2 \cdot 2 = -4 \text{ (infeasible).}$$

Case 4: $\mu_1 \neq 0, \mu_2 \neq 0 \implies x = 2, y = 3$

$$\mu_1 = -2 \cdot 2 = -4, \quad \mu_2 = -2 \cdot 3 = -6 \text{ (infeasible).}$$

Solved Problem 3: Solution

Step 3: Solve KKT Conditions From stationarity: $\mu_1 = -2x$, $\mu_2 = -2y$.
Case 1: $\mu_1 = 0, \mu_2 = 0$

$$x = 0, \quad y = 0$$

Check: $x - 2 = 0 - 2 = -2 \leq 0$, $y - 3 = 0 - 3 = -3 \leq 0$ (feasible). Objective: $f(0,0) = 0$.
Case 2: $\mu_1 = 0, \mu_2 \neq 0 \implies y = 3$

$$x = 0, \quad y = 3, \quad \mu_2 = -2 \cdot 3 = -6 \text{ (infeasible, since } \mu_2 \geq 0).$$

Case 3: $\mu_1 \neq 0, \mu_2 = 0 \implies x = 2$

$$x = 2, \quad y = 0, \quad \mu_1 = -2 \cdot 2 = -4 \text{ (infeasible)}.$$

Case 4: $\mu_1 \neq 0, \mu_2 \neq 0 \implies x = 2, y = 3$

$$\mu_1 = -2 \cdot 2 = -4, \quad \mu_2 = -2 \cdot 3 = -6 \text{ (infeasible)}.$$

Step 4: Classify Only feasible solution: $(0,0)$, $f = 0$. Hessian: $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, positive definite. **Solution**: Minimum at $(x,y) = (0,0)$, $f = 0$.

Solved Problem 4: Linear Objective

Maximize $f(x, y) = 3x + 4y$ subject to:

$$g(x, y) = x^2 + y^2 - 25 \leq 0$$

Step 1: Form Lagrangian

$$\mathcal{L}(x, y, \mu) = 3x + 4y + \mu(x^2 + y^2 - 25)$$

Step 2: Write KKT Conditions

$$\text{Stationarity: } \frac{\partial \mathcal{L}}{\partial x} = 3 + 2\mu x = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = 4 + 2\mu y = 0$$

$$\text{Primal Feasibility: } x^2 + y^2 - 25 \leq 0$$

$$\text{Dual Feasibility: } \mu \geq 0$$

$$\text{Complementary Slackness: } \mu(x^2 + y^2 - 25) = 0$$

Solved Problem 4: Solution

Step 3: Solve KKT Conditions From stationarity: $x = -\frac{3}{2\mu}$, $y = -\frac{4}{2\mu} = -\frac{2}{\mu}$.

Solved Problem 4: Solution

Step 3: Solve KKT Conditions From stationarity: $x = -\frac{3}{2\mu}$, $y = -\frac{4}{2\mu} = -\frac{2}{\mu}$.
Case 1: $\mu = 0$

$$x = \text{undefined}, \quad y = \text{undefined (infeasible)}.$$

Solved Problem 4: Solution

Step 3: Solve KKT Conditions From stationarity: $x = -\frac{3}{2\mu}$, $y = -\frac{4}{2\mu} = -\frac{2}{\mu}$.
Case 1: $\mu = 0$

$$x = \text{undefined}, \quad y = \text{undefined (infeasible)}.$$

Case 2: $\mu \neq 0 \implies x^2 + y^2 = 25$

$$\left(-\frac{3}{2\mu}\right)^2 + \left(-\frac{2}{\mu}\right)^2 = 25 \implies \frac{9}{4\mu^2} + \frac{4}{\mu^2} = 25 \implies \frac{9+16}{4\mu^2} = 25 \implies \frac{25}{4\mu^2} = 25 \implies \mu^2 =$$

$$x = -\frac{3}{2 \cdot \frac{1}{2}} = -3, \quad y = -\frac{2}{\frac{1}{2}} = -4$$

Check: $x^2 + y^2 = (-3)^2 + (-4)^2 = 9 + 16 = 25 \leq 25$ (feasible). Objective: $f(-3, -4) = 3(-3) + 4(-4) = -9 - 16 = -25$.

Solved Problem 4: Solution

Step 3: Solve KKT Conditions From stationarity: $x = -\frac{3}{2\mu}$, $y = -\frac{4}{2\mu} = -\frac{2}{\mu}$.
Case 1: $\mu = 0$

$$x = \text{undefined}, \quad y = \text{undefined (infeasible)}.$$

Case 2: $\mu \neq 0 \implies x^2 + y^2 = 25$

$$\left(-\frac{3}{2\mu}\right)^2 + \left(-\frac{2}{\mu}\right)^2 = 25 \implies \frac{9}{4\mu^2} + \frac{4}{\mu^2} = 25 \implies \frac{9+16}{4\mu^2} = 25 \implies \frac{25}{4\mu^2} = 25 \implies \mu^2 =$$

$$x = -\frac{3}{2 \cdot \frac{1}{2}} = -3, \quad y = -\frac{2}{\frac{1}{2}} = -4$$

Check: $x^2 + y^2 = (-3)^2 + (-4)^2 = 9 + 16 = 25 \leq 25$ (feasible). Objective: $f(-3,-4) = 3(-3) + 4(-4) = -9 - 16 = -25$.
Step 4: Classify For maximization, check boundary points or alternative points. Test: $x = 3, y = 4$:

$$3^2 + 4^2 = 25, \quad f(3,4) = 3 \cdot 3 + 4 \cdot 4 = 9 + 16 = 25$$

Solution: Maximum at $(x, y) = (3, 4)$, $f = 25$.

Solved Problem 5: Three Variables

Minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to:

$$g(x, y, z) = x + y + z - 6 \leq 0$$

Step 1: Form Lagrangian

$$\mathcal{L}(x, y, z, \mu) = x^2 + y^2 + z^2 + \mu(x + y + z - 6)$$

Step 2: Write KKT Conditions

Stationarity: $\frac{\partial \mathcal{L}}{\partial x} = 2x + \mu = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = 2y + \mu = 0, \quad \frac{\partial \mathcal{L}}{\partial z} = 2z + \mu = 0$

Primal Feasibility: $x + y + z - 6 \leq 0$

Dual Feasibility: $\mu \geq 0$

Complementary Slackness: $\mu(x + y + z - 6) = 0$

Solved Problem 5: Solution

Step 3: Solve KKT Conditions From stationarity:
$\mu = -2x = -2y = -2z \implies x = y = z.$

Solved Problem 5: Solution

Step 3: Solve KKT Conditions From stationarity:
$\mu = -2x = -2y = -2z \implies x = y = z$.
Case 1: $\mu = 0$

$$x = 0, \quad y = 0, \quad z = 0$$

Check: $0 + 0 + 0 = 0 \leq 6$ (feasible). Objective: $f(0,0,0) = 0$.

Solved Problem 5: Solution

Step 3: Solve KKT Conditions From stationarity:
$\mu = -2x = -2y = -2z \implies x = y = z$.
Case 1: $\mu = 0$

$$x = 0, \quad y = 0, \quad z = 0$$

Check: $0 + 0 + 0 = 0 \leq 6$ (feasible). Objective: $f(0,0,0) = 0$.
Case 2: $\mu \neq 0 \implies x + y + z = 6$

$$x = y = z, \quad x + x + x = 6 \implies 3x = 6 \implies x = 2, \quad y = 2, \quad z = 2$$

$$\mu = -2 \cdot 2 = -4 \text{ (infeasible, since } \mu \geq 0\text{)}.$$

Solved Problem 5: Solution

Step 3: Solve KKT Conditions From stationarity:
$\mu = -2x = -2y = -2z \implies x = y = z$.
Case 1: $\mu = 0$

$$x = 0, \quad y = 0, \quad z = 0$$

Check: $0 + 0 + 0 = 0 \leq 6$ (feasible). Objective: $f(0,0,0) = 0$.
Case 2: $\mu \neq 0 \implies x + y + z = 6$

$$x = y = z, \quad x + x + x = 6 \implies 3x = 6 \implies x = 2, \quad y = 2, \quad z = 2$$

$$\mu = -2 \cdot 2 = -4 \text{ (infeasible, since } \mu \geq 0\text{)}.$$

Step 4: Classify Hessian: $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, positive definite. Only feasible solution: $(0,0,0)$, $f = 0$. **Solution**: Minimum at $(x, y, z) = (0,0,0)$, $f = 0$.

Solved Problem 6: Nonlinear Constraint

Maximize $f(x, y) = xy$ subject to:

$$g(x, y) = x^2 + y^2 - 4 \leq 0$$

Step 1: Form Lagrangian

$$\mathcal{L}(x, y, \mu) = xy + \mu(x^2 + y^2 - 4)$$

Step 2: Write KKT Conditions

Stationarity: $\dfrac{\partial \mathcal{L}}{\partial x} = y + 2\mu x = 0, \quad \dfrac{\partial \mathcal{L}}{\partial y} = x + 2\mu y = 0$

Primal Feasibility: $x^2 + y^2 - 4 \leq 0$

Dual Feasibility: $\mu \geq 0$

Complementary Slackness: $\mu(x^2 + y^2 - 4) = 0$

Solved Problem 6: Solution

Step 3: Solve KKT Conditions From stationarity: $y = -2\mu x$, $x = -2\mu y$.

Solved Problem 6: Solution

Step 3: Solve KKT Conditions From stationarity: $y = -2\mu x$, $x = -2\mu y$.
Case 1: $\mu = 0$

$$y = 0, \quad x = 0$$

Check: $0^2 + 0^2 = 0 \leq 4$ (feasible). Objective: $f(0,0) = 0$.

Solved Problem 6: Solution

Step 3: Solve KKT Conditions From stationarity: $y = -2\mu x$, $x = -2\mu y$.
Case 1: $\mu = 0$

$$y = 0, \quad x = 0$$

Check: $0^2 + 0^2 = 0 \leq 4$ (feasible). Objective: $f(0,0) = 0$.
Case 2: $\mu \neq 0 \implies x^2 + y^2 = 4$

$$y = -2\mu x, \quad x = -2\mu y \implies y = -2\mu(-2\mu y) = 4\mu^2 y \implies y(4\mu^2 - 1) = 0$$

If $y = 0$, then $x = 0$, contradicts $x^2 + y^2 = 4$. So, $4\mu^2 - 1 = 0 \implies \mu = \frac{1}{2}$.

$$y = -2 \cdot \frac{1}{2} x = -x, \quad x^2 + (-x)^2 = 4 \implies x^2 + x^2 = 4 \implies 2x^2 = 4 \implies x = \pm\sqrt{2}, \quad y = \mp$$

Check: $x^2 + y^2 = (\sqrt{2})^2 + (-\sqrt{2})^2 = 2 + 2 = 4$. Objective:
$f(\sqrt{2}, -\sqrt{2}) = \sqrt{2} \cdot (-\sqrt{2}) = -2$, $f(-\sqrt{2}, \sqrt{2}) = -2$. Test: $f(\sqrt{2}, \sqrt{2}) = 2$ (feasible, since $(\sqrt{2})^2 + (\sqrt{2})^2 = 4$).

Solved Problem 6: Solution

Step 3: Solve KKT Conditions From stationarity: $y = -2\mu x$, $x = -2\mu y$.
Case 1: $\mu = 0$

$$y = 0, \quad x = 0$$

Check: $0^2 + 0^2 = 0 \leq 4$ (feasible). Objective: $f(0,0) = 0$.
Case 2: $\mu \neq 0 \implies x^2 + y^2 = 4$

$$y = -2\mu x, \quad x = -2\mu y \implies y = -2\mu(-2\mu y) = 4\mu^2 y \implies y(4\mu^2 - 1) = 0$$

If $y = 0$, then $x = 0$, contradicts $x^2 + y^2 = 4$. So, $4\mu^2 - 1 = 0 \implies \mu = \frac{1}{2}$.

$$y = -2 \cdot \frac{1}{2} x = -x, \quad x^2 + (-x)^2 = 4 \implies x^2 + x^2 = 4 \implies 2x^2 = 4 \implies x = \pm\sqrt{2}, \quad y = \mp$$

Check: $x^2 + y^2 = (\sqrt{2})^2 + (-\sqrt{2})^2 = 2 + 2 = 4$. Objective:
$f(\sqrt{2}, -\sqrt{2}) = \sqrt{2} \cdot (-\sqrt{2}) = -2$, $f(-\sqrt{2}, \sqrt{2}) = -2$. Test: $f(\sqrt{2}, \sqrt{2}) = 2$ (feasible, since $(\sqrt{2})^2 + (\sqrt{2})^2 = 4$).
Step 4: Classify Maximum at $(x, y) = (\sqrt{2}, \sqrt{2})$ or $(-\sqrt{2}, -\sqrt{2})$, $f = 2$.

Solved Problem 7: Support Vector Machine Margin

Maximize $f(w_1, w_2) = \frac{1}{w_1^2 + w_2^2}$ (maximize margin) subject to:

$$g(w_1, w_2) = w_1 + w_2 - 2 \leq 0$$

Equivalently, minimize $f(w_1, w_2) = w_1^2 + w_2^2$. **Step 1: Form Lagrangian**

$$\mathcal{L}(w_1, w_2, \mu) = w_1^2 + w_2^2 + \mu(w_1 + w_2 - 2)$$

Step 2: Write KKT Conditions

$$\text{Stationarity: } \frac{\partial \mathcal{L}}{\partial w_1} = 2w_1 + \mu = 0, \quad \frac{\partial \mathcal{L}}{\partial w_2} = 2w_2 + \mu = 0$$

$$\text{Primal Feasibility: } w_1 + w_2 - 2 \leq 0$$

$$\text{Dual Feasibility: } \mu \geq 0$$

$$\text{Complementary Slackness: } \mu(w_1 + w_2 - 2) = 0$$

Solved Problem 7: Solution

Step 3: Solve KKT Conditions From stationarity: $\mu = -2w_1 = -2w_2 \implies w_1 = w_2$.

Solved Problem 7: Solution

Step 3: Solve KKT Conditions From stationarity: $\mu = -2w_1 = -2w_2 \implies w_1 = w_2$.
Case 1: $\mu = 0$

$$w_1 = 0, \quad w_2 = 0$$

Check: $0 + 0 = 0 \leq 2$ (feasible). Objective: $f(0,0) = 0$ (undefined for original $\frac{1}{w_1^2 + w_2^2}$).

Solved Problem 7: Solution

Step 3: Solve KKT Conditions From stationarity: $\mu = -2w_1 = -2w_2 \implies w_1 = w_2$.
Case 1: $\mu = 0$

$$w_1 = 0, \quad w_2 = 0$$

Check: $0+0=0 \leq 2$ (feasible). Objective: $f(0,0) = 0$ (undefined for original $\frac{1}{w_1^2+w_2^2}$).
Case 2: $\mu \neq 0 \implies w_1 + w_2 = 2$

$$w_1 = w_2, \quad w_1 + w_1 = 2 \implies w_1 = 1, \quad w_2 = 1$$

$$\mu = -2 \cdot 1 = -2 \text{ (infeasible, since } \mu \geq 0).$$

Solved Problem 7: Solution

Step 3: Solve KKT Conditions From stationarity: $\mu = -2w_1 = -2w_2 \implies w_1 = w_2$.
Case 1: $\mu = 0$

$$w_1 = 0, \quad w_2 = 0$$

Check: $0 + 0 = 0 \leq 2$ (feasible). Objective: $f(0,0) = 0$ (undefined for original $\frac{1}{w_1^2 + w_2^2}$).
Case 2: $\mu \neq 0 \implies w_1 + w_2 = 2$

$$w_1 = w_2, \quad w_1 + w_1 = 2 \implies w_1 = 1, \quad w_2 = 1$$

$$\mu = -2 \cdot 1 = -2 \text{ (infeasible, since } \mu \geq 0).$$

Step 4: Classify Hessian: $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, positive definite. No feasible interior solution.
Check boundary: $w_1 = 1, w_2 = 1$, but infeasible μ. Test boundary points: $f(1,1) = 1 + 1 = 2$, $w_1 + w_2 = 2$. **Solution**: Minimum at $(w_1, w_2) = (0,0)$, but for SVM, boundary solution $(1,1)$ may be considered with $f = 2$.

Application: Resource Allocation

Maximize $f(x, y) = 30x + 50y - 2x^2 - 2y$ subject to:

$$g(x, y) = x + y - 10 \leq 0$$

Use the Lagrangean method and KKT conditions to solve.

Interpret the solution in the context of resource allocation (e.g., labor and capital).

Analyze the impact of tightening the constraint to $x + y \leq 8$.

Exercise 1

Minimize $f(x, y) = x^2 + 2y^2$ subject to:

$$x + y - 5 \leq 0$$

Use KKT conditions to find the optimal solution.

Verify feasibility and classify the solution.

Discuss whether the constraint is active.

Exercise 2

Maximize $f(x, y) = xy$ subject to:

$$x^2 + y^2 - 16 \leq 0$$

Solve using KKT conditions.

Determine if the solution lies on the constraint boundary.

Interpret the solution geometrically.

Assignment 1 (Individual)

Minimize $f(x, y) = (x - 2)^2 + (y - 3)^2$ subject to:

$$x + y - 6 \leq 0$$

Solve using KKT conditions.

Submit a report including:

Detailed KKT solution steps.
Verification of feasibility and optimality.
Interpretation of the solution.

Assignment 2 (Team Work)

Minimize $f(w_1, w_2) = w_1^2 + 2w_2^2 + w_1 w_2$ subject to:

$$w_1 + w_2 - 4 \leq 0$$

- Form a team of 3–4 students.
- Solve using KKT conditions.
- Prepare a presentation including:
  - Solution process and verification.
  - Interpretation in neural network training.
  - Sensitivity to constraint changes.
- Deadline: One week from today.

References

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NPTEL: Prof. S. S. Rao, https://onlinecourses.nptel.ac.in/noc20_me46/preview

Thank You

Questions?